

Jonas Ngnawé

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Education

Université Laval, Mila - Quebec AI Institute, Stanford University (VSR) <i>PhD in Computer Science: Adversarial Machine Learning</i>	Quebec City, Canada 2021–2027 (expected)
AIMS African Master's of Machine Intelligence <i>MSc in Machine Learning</i> Funded by Meta and Google; taught by world top researchers.	Kigali, Rwanda 2018–2019
African Institute for Mathematical Sciences (AIMS) <i>MSc in Mathematical Sciences</i>	Limbe, Cameroon 2016–2017
École Polytechnique Yaoundé <i>Engineering Degree (MSc equivalent), Computer Engineering</i> Five-year integrated engineering programme.	Yaoundé, Cameroon 2011–2016
Summer Schools: Machine Learning Summer School (MLSS), London 2019 ; Deep Learning & Reinforcement Learning Summer School (DLRL), Edmonton 2025 .	

Publications

Peer-reviewed conference papers.....

- Detecting Brittle Decisions for Free: Leveraging Margin Consistency in Deep Robust Classifiers. **Ngnawé, J., Sahoo, S., Pequignot, Y., Précioso, F., & Gagné, C. NeurIPS 2024.** [Proceedings]
- Robust Fine-Tuning from Non-Robust Pretrained Models: Mitigating Suboptimal Transfer with Epsilon-Scheduling. **Ngnawé, J., Heuillet, M., Sahoo, S., Pequignot, Y., Ahmad, O., Durand, A., Précioso, F., & Gagné, C. ICLR 2026** (earlier version: NeurIPS 2025 Workshop on Reliable ML). [OpenReview]
- A Layer Selection Approach to Test Time Adaptation. **Sahoo, S., ElAraby, M., Ngnawé, J., Pequignot, Y., Précioso, F., & Gagné, C. AAAI 2025.** [Proceedings]

Workshop papers.....

- A Guide to Robust Generalization: The Impact of Architecture, Pre-training, and Optimization Strategy. **Heuillet, M., Bhagwatkar, R., Ngnawé, J., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A. NeurIPS 2025 Workshop on Reliable ML.** [arXiv]
- Robustmix: Improving Robustness by Regularizing the Frequency Bias of Deep Networks. **Ngnawé, J., Njifon, M. A., Heek, J., & Dauphin, Y. NeurIPS 2022 Workshop on Distribution Shifts.** [OpenReview]

Under review.....

- RobustGenBench: A Benchmark for Robust Generalization to Adversarial and Common Perturbations, with Applications to Vision and Vision-Enabled Large Language Models. **Heuillet, M., Ngnawé, J., Bhagwatkar, R., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A.**

Master's projects.....

- Scalable Inference with Distributed Gaussian Processes using the Wasserstein Barycenter. **Ngnawé, J., Deisenroth, M.** [GitHub]
- k-Clustering Problem: Application to a Field Technician Allocation Optimization Problem. **Ngnawé, J., Zacko, E., Wandji, N., & Ndjinga, M.** [ResearchGate]

AI/ML Work Experience

Vector Institute (FastLane MLA Program) <i>Machine Learning Associate</i>	Toronto, Canada May – Aug 2026
○ Optimized LLM inference for AutoInference.ai: vLLM serving config to increase goodput and reducing latency.	

Stanford Artificial Intelligence Lab (SAIL)

Visiting Student Researcher

Palo Alto, USA

Oct 2025 – June 2026

- Safety, robustness and uncertainty estimation in multimodal vision–language models.

Google Brain

Google AI Resident

Accra, Ghana

Oct 2019 – Sep 2021

- Developed robust training methods for deep networks; frequency-bias regularization work published at the NeurIPS 2022 Distribution Shifts workshop.
- Compressed deep networks via quantization and pruning: Mixed-precision quantization ($\{4, 8\}$ -bit quantization, with 0-bit for sparsity to retain high accuracy on vision models).

Earlier Software Engineering Experience

GroupOne Holding Company

Computer Engineer

Limbe, Cameroon

Jan – Aug 2018

Web development and fuel-consumption prediction; industrial maintenance of Honeywell Experion refinery systems. *JavaScript, PHP, MySQL*.

Freelance

Software Developer

Cameroon

2017–2018

Digitized administrative procedures for Cameroonian ministries (Maarch RM electronic document management); built and maintained WooCommerce plugins. *PHP, Java, MySQL, Git*.

Bolloré Africa Logistics

Software Developer Intern

Douala, Cameroon

Feb – Aug 2016

Automated review of information-system access rights and built anomaly detection for abnormal privileges. *Talend, QlikView, C#, ASP.NET, SQL Server*.

Talks

Host, Kmer AI Talks.

2024: Panelist, “AI Regulation & Fairness in the Generative AI Era,” Black in AI Workshop @ NeurIPS.

2024–2025: “Detecting Brittle Decisions for Free” — Kmer AI (Jun 2024); Galsen AI (Jan 2025).

2022: Keynote, “Robustness in AI,” Deep Learning IndabaX-Cameroon (Dec 2022).

Service

Reviewer: NeurIPS, ICLR, AAAI; IEEE Transactions on Image Processing.

Algorithmic Fairness workshops @ NeurIPS 2021–2024; Montreal AI Symposium (2022, 2024); Black in AI Workshop (2022–2024); Deep Learning Indaba / Pan-African ML Conference (2023).

Organizing: Interpolate Workshop @ NeurIPS 2022 (First Workshop on Interpolation Regularizers and Beyond), organizing committee.

Co-organizer, Machine Learning Projects Competition (MLPC) 2020–2022, Polytechnique Yaoundé, \$2,000 sponsorship.

Co-organizer: two-week introductory ML course (Sep 2019), Polytechnique Yaoundé; first Deep Learning IndabaX Rwanda (Apr 2019), Kigali.

Co-founder, Kmer-AI — association promoting AI and ML in Cameroon (since 2020).

AI/ML Skills

Frameworks: PyTorch, TensorFlow, Scikit-Learn

Infrastructure: Git, Slurm, Skypilot, vLLM, Alperfi, Docker, Weights & Biases, GCP, AWS

Research Related: Adversarial robustness, uncertainty estimation, training and finetuning Large VLMs, SFT/RL